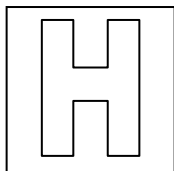


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PIONEER JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION
HIGHER 1

CHEMISTRY

8872/01

Paper 1 Multiple Choice

20 September 2012

50 min

Additional Materials: Multiple Choice Answer Sheet
 Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

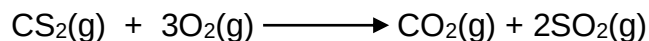
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Section A

For each question, there are four possible answers labelled **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

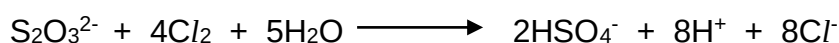
- 1 Carbon disulfide, CS_2 , is a volatile flammable liquid used in the manufacture of cellophane. On combustion, CS_2 is oxidised as follows.



A 20 cm^3 sample of carbon disulfide vapour is ignited with 100 cm^3 of oxygen. The final volume of gases after burning is treated with an excess of aqueous sodium hydroxide.

What percentage of this final volume dissolved in aqueous sodium hydroxide?

- A** 20 %
B 40 %
C 60 %
D 100 %
- 2 Sodium thiosulfate is used in the textile industry to remove an excess of chlorine from bleaching processes by reducing it to chloride ions.

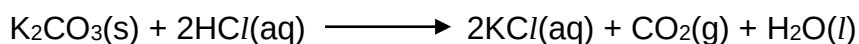


In this reaction, how many moles of electrons are supplied per mole of thiosulfate?

- A** 2 **B** 4 **C** 6 **D** 8
- 3 Which of the following is **not** true about P^{3-} ion?
- A** It is very polarisable.
B It is isoelectronic with argon.
C Its p-orbitals are completely filled.
D It is smaller than a phosphorus atom.

- 4 In which of the following pairs is the bond angle of the second molecule / ion greater than that of the first?
- A NO_2^+ , NO_2^-
- B SO_4^{2-} , SO_3^{2-}
- C SO_3^{2-} , CO_3^{2-}
- D AlCl_3 , PCl_3
- 5 Which one of the following **cannot** be explained by hydrogen bonds?
- A CH_3CHO has a higher boiling point than C_2H_6 .
- B CH_3COOH has a M_r of 120 in benzene.
- C HF has a higher boiling point than HCl.
- D CH_3COCH_3 is miscible in water.
- 6 In an experiment, 1.00 g of solid K_2CO_3 was added to a polystyrene cup containing 25.0 cm^3 of 1.0 mol dm^{-3} aqueous HCl. It was found that the temperature of the solution rose by 2.1°C .

The equation for the reaction is as follows.



What is the correct value for the enthalpy change of reaction above?
(Assume that the heat capacity of the solution is $4.2 \text{ J K}^{-1} \text{ cm}^{-3}$)

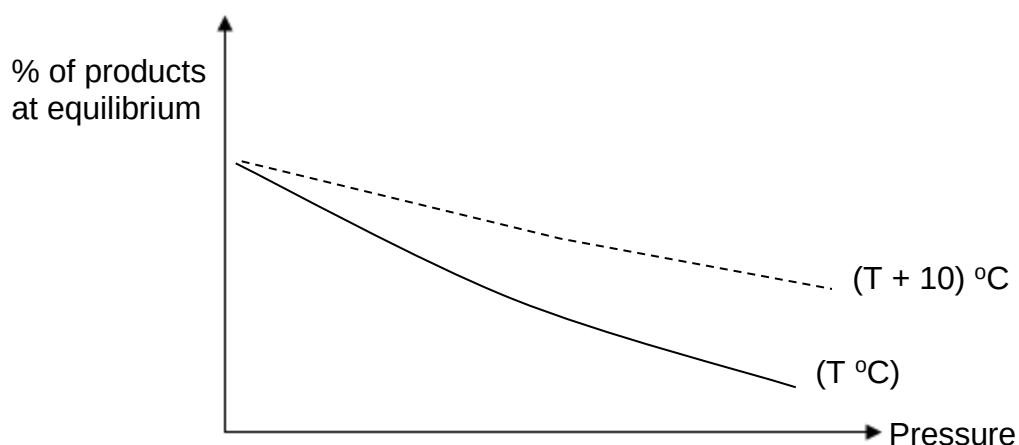
- A $-8.82 \text{ kJ mol}^{-1}$
- B $+8.82 \text{ kJ mol}^{-1}$
- C $-30.5 \text{ kJ mol}^{-1}$
- D $+30.5 \text{ kJ mol}^{-1}$

- 7 In an experiment, 80 cm³ of water at 25°C was brought to boiling point by burning propane in excess of oxygen. Given that the enthalpy change of combustion of propane is -2220 kJ mol⁻¹, calculate the volume of propane needed if this process is only 90% efficient.

Assume that the heat capacity of water is 4.3 J K⁻¹ cm⁻³ and the experiment is done under room temperature and pressure.

- A 103 cm³ B 251 cm³ C 279 cm³ D 310 cm³

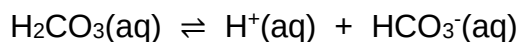
- 8 The graph below shows the variation of the percentage of gaseous products present at equilibrium with temperature and pressure for a particular reaction.



Which one of the following systems could the graphs represent?

- A $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ $\Delta H = -92 \text{ kJ mol}^{-1}$
- B $3\text{O}_2(\text{g}) + 4\text{NH}_3(\text{g}) \rightleftharpoons 2\text{N}_2(\text{g}) + 6\text{H}_2\text{O}(\text{g})$ $\Delta H = -1248 \text{ kJ mol}^{-1}$
- C $2\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{N}_2\text{O}(\text{g})$ $\Delta H = +82 \text{ kJ mol}^{-1}$
- D $\text{CO}_2(\text{g}) + \text{C}(\text{s}) \rightleftharpoons 2\text{CO}(\text{g})$ $\Delta H = +173 \text{ kJ mol}^{-1}$

- 9 Human plasma is buffered mainly by dissolved CO_2 which has reacted to form carbonic acid.



Usually the pH of human plasma is 7.4. The acid dissociation constant, K_a , of carbonic acid is $7.90 \times 10^{-7} \text{ mol dm}^{-3}$. Which one of the following statements is correct?

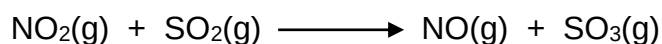
- A The ratio of $[\text{HCO}_3^-]$ to $[\text{H}_2\text{CO}_3]$ in human plasma is 1 : 20
 - B The ratio of $[\text{HCO}_3^-]$ to $[\text{H}_2\text{CO}_3]$ in human plasma is 20 : 1
 - C This buffer system is more efficient in removing excess base than excess acid.
 - D This buffer system can be prepared by mixing suitable amounts of sodium hydrogencarbonate and sodium hydroxide.
- 10 The pH of water varies with temperature according to the table below.

Temperature / °C	pH
25	7.0
62	6.5

Which of the following statements is correct?

- A Water is not a neutral liquid at 62°C.
- B The concentration of H^+ ions is higher than the concentration of OH^- ions in water at 62°C.
- C The ionic product of water, K_w , decreases at higher temperature.
- D The ionic dissociation of water is an endothermic process.

- 11 The reaction between NO_2 and SO_2 is a key stage in the formation of acid rain.



The initial rate of this reaction was measured for different concentrations of reactants and the following results were obtained.

$[\text{NO}_2] / \text{mol dm}^{-3}$	$[\text{SO}_2] / \text{mol dm}^{-3}$	Relative rate
0.005	0.005	1.00
0.006	0.006	1.44
0.008	0.006	1.92

Which one of the following gives the correct orders of reaction with respect to the two reactants?

	Order with respect to $[\text{NO}_2]$	Order with respect to $[\text{SO}_2]$
A	1	0
B	1	1
C	2	0
D	2	1

- 12 A chemical plant illegally dumped some radioactive waste in a landfill. This waste contains of two radioactive isotopes **X** and **Y** in the proportion 4:1. The decay of the radioactive isotopes follow first order kinetics. The half-life of **X** is 2 days whereas that of **Y** is 4 days. By the time the authorities found out about this illegal dumping, there were equal amounts of **X** and **Y**.

How long was the waste in the landfill before the authorities found out?

- A 2 days B 4 days C 8 days D 16 days

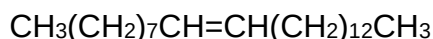
- 13 Which of the following oxide will not dissolve in sodium hydroxide?

- A MgO
 B SiO_2
 C P_4O_{10}
 D Al_2O_3

- 14 Element **E** is in the third period of the Period Table. The chloride of **E** has a simple molecular structure, while the oxide of **E** has a giant ionic lattice structure. Which one of the following statements is **incorrect**?
- A The atomic radius of **E** is larger than that of chlorine.
 - B** The second ionisation energy of **E** is higher than that of sodium.
 - C The chloride of **E** dissolves in water to give an acidic solution.
 - D The oxide of **E** reacts with excess aqueous sodium hydroxide to form a colourless solution.
- 15 Which one of the following alcohols, upon dehydration with hot concentrated sulfuric acid, produces one and only one alkene that displays cis-trans isomers?
- A pentan-1-ol
 - B pentan-2-ol
 - C** pentan-3-ol
 - D butan-2-ol
- 16 How many structural isomers with molecular formula $C_5H_{10}O$ give an orange precipitate with 2,4-dinitrophenylhydrazine but no precipitate with alkaline aqueous iodine?
- A 3
 - B 4
 - C** 5
 - D 6

- 17 Fly paper is used as a non-toxic method of trapping houseflies. To increase its effectiveness and attractiveness, Muscalure, which is a fly sex pheromone, is added to the paper during its manufacture.

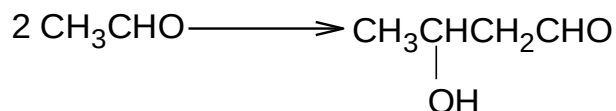
Muscalure has the following structure:



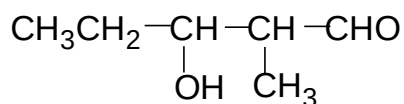
Which of the following statements about Muscalure is **incorrect**?

- A** It gives diol with cold, dilute potassium dichromate (VI).
- B** It exists as a pair of geometric isomers.
- C** In the presence of excess bromine and uv light, it undergoes substitution.
- D** It can be extracted from the fly paper by soaking the paper in benzene.
- 18 The length of a carbon-carbon single bond is 0.154 nm, whereas that of the carbon-carbon double bond is 0.134 nm. What is the likely carbon-carbon bond length between adjacent carbon atoms in benzene?
- A** 0.134 nm
- B** 0.139 nm
- C** 0.154 nm
- D** 0.154 nm for three bonds and 0.134 nm for the remaining three bonds.
- 19 Which of the following cannot be synthesised from propan-1-ol within 2 steps or less?
- A** ethanoic acid
- B** propanoic acid
- C** ethanol
- D** propan-2-ol

- 20 In the presence of a dilute alkali, some aldehydes and ketones undergo the 'adol reaction', where they dimerise to form hydroxylcarbonyl compounds (an aldol). For example, ethanal dimerises in this way to form 3-hydroxybutanal.



Which one of the following carbonyl compounds will undergo aldol reaction to produce the aldol shown below?



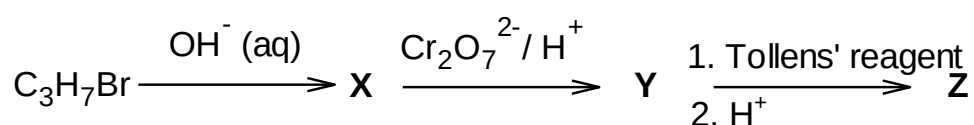
- A** $\text{CH}_3\text{CH}_2\text{CHO}$
- B** $\text{CH}_3\text{CH}(\text{CH}_3)\text{CHO}$
- C** CH_3COCH_3
- D** $\text{CH}_3\text{COCH}_2\text{CH}_3$
- 21 Which one of the following statements is true about the product formed when ethanol is distilled with acidified aqueous potassium dichromate (VI)?
- A** It can be reduced to ethanoic acid.
- B** It can undergo reaction with PCl_5 .
- C** It can undergo addition reaction.
- D** It would not give a precipitate with Fehling's solution.
- 22 Which sequence shows the correct order of increasing value of pK_a .
- A** $\text{CH}_3\text{CH}_2\text{OH}$, $\text{ClCH}_2\text{CH}_2\text{OH}$, CH_3COOH
- B** $\text{CH}_3\text{CF}_2\text{COOH}$, $\text{FCH}_2\text{CHFCOOH}$, $\text{F}_2\text{CHCH}_2\text{COOH}$
- C** $\text{CH}_3\text{CH}_2\text{COOH}$, $\text{ClCH}_2\text{CH}_2\text{COOH}$, $\text{CH}_3\text{CHClCOOH}$
- D** BrCH_2COOH , ClCH_2COOH , FCH_2COOH

- 23 Compound **N** was refluxed with aqueous sodium hydroxide and the resulting mixture was distilled. The distillate gave a positive tri-iodomethane test; the residue in the distillation flask, after acidification, gave a white precipitate.

Which of the following could be **N**?

- A $\text{C}_6\text{H}_5\text{COOC}(\text{CH}_3)_3$
 B $\text{CH}_3\text{CH}_2\text{COOC}_6\text{H}_5$
 C $\text{CH}_3\text{CH}_2\text{OCOC}_6\text{H}_5$
 D $\text{C}_6\text{H}_5\text{OCOC}(\text{CH}_3)_3$

- 24 The compound $\text{C}_3\text{H}_7\text{Br}$ undergoes a sequence of reactions as follows:



What of the following shows the correct identities of **X**, **Y** and **Z**?

- | X | Y | Z |
|--|---|-------------------------------------|
| A $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ | $\text{CH}_3\text{CH}_2\text{CHO}$ | $\text{CH}_3\text{CH}_2\text{COOH}$ |
| B $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ | $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{OH}$ | CH_3COOH |
| C $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ | $\text{CH}_3\text{CH}_2\text{COOH}$ | $\text{CH}_3\text{CH}_2\text{CHO}$ |
| D $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$ | CH_3COCH_3 | CH_3COOH |

- 25 The same carboxylic acid is obtained either by a hydrolysis of nitrile **P** or by the oxidation of alcohol **Q**.

Which of the following pairs could be **P** and **Q**?

- | P | Q |
|--|---|
| A $\text{CH}_3\text{CH}_2\text{CN}$ | $\text{CH}_3\text{CH}_2\text{OH}$ |
| B $\text{C}_6\text{H}_5\text{CH}_2\text{CN}$ | $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$ |
| C $(\text{CH}_3)_2\text{CHCN}$ | $(\text{CH}_3)_3\text{COH}$ |
| D $\text{C}_6\text{H}_5\text{CH}_2\text{CN}$ | $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{OH}$ |

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

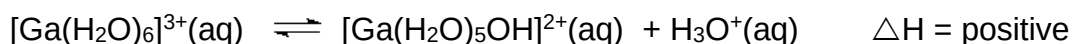
Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements which you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

26 The gallium hydrate hydrolyses as shown below.

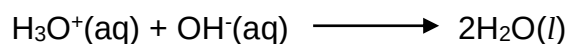


Which of the following statements are correct?

- 1** $[\text{Ga}(\text{H}_2\text{O})_6]^{3+}$ is more stable at low pH values.
- 2** Increasing the temperature will favour the formation of $[\text{Ga}(\text{H}_2\text{O})_5\text{OH}]^{2+}$.
- 3** Increasing the concentration of $[\text{Ga}(\text{H}_2\text{O})_6]^{3+}$ will increase K_c as the forward reaction is favoured.

27 Which statements are correct for the neutralisation of a strong acid by a strong alkali in aqueous solution at 25 °C ?

- 1** It can be represented as



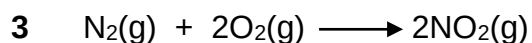
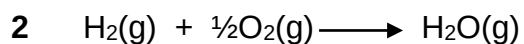
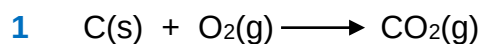
- 2** The enthalpy change per mole of H_2O formed is independent of the acid or alkali used.
- 3** It is an endothermic process.

The responses **A** to **D** should be selected on the basis of

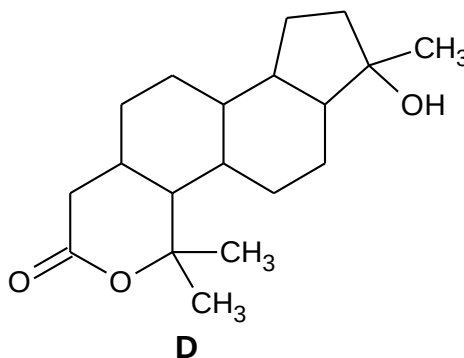
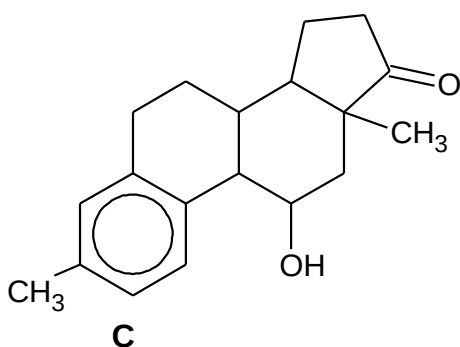
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

28 For which of the following reactions does the value of ΔH^θ represent both a standard enthalpy change of combustion and a standard enthalpy change of formation?



29 The structures of compounds **C** and **D** are shown below.



Which of the following reagents can be used to distinguish **C** and **D**?

1 2,4-dinitrophenylhydrazine

2 hot acidified potassium manganate(VII) solution

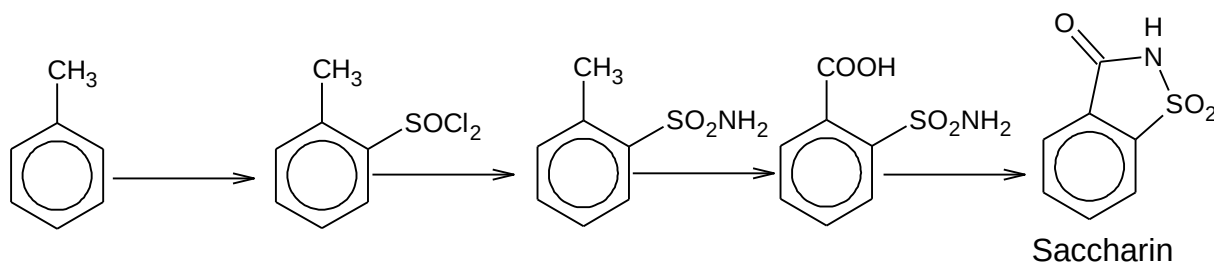
3 alkaline aqueous iodine

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 30** Saccharin is an artificial sweetening agent used in some soft drinks and is manufactured from methylbenzene by the following series of reactions.



What types of reactions are involved in the series?

- 1** condensation
- 2** substitution
- 3** oxidation

- End Of Paper -

PIONEER JUNIOR COLLEGE
JC 2 H1 Prelim 2012

CHEMISTRY H1

8872/01

PAPER 1

MCQ Answers

1	C	11	B	21	C
2	D	12	C	22	B
3	D	13	A	23	C
4	C	14	B	24	A
5	A	15	C	25	D
6	C	16	C	26	B
7	D	17	A	27	B
8	D	18	B	28	D
9	B	19	C	29	B
10	D	20	A	30	A